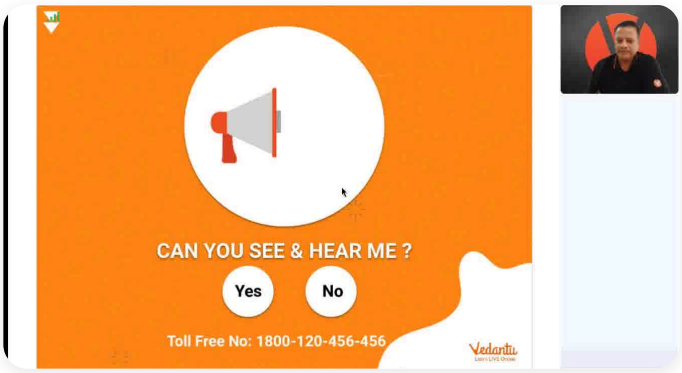
Notes

Welcome and Introduction to JEE Droppers Batch



Welcome to the Session

Welcome to the second session of the JEE Droppers Batch for Mathematics. This batch focuses on Mathematics 2, which includes Algebra and Coordinate Geometry. The instructor extends best wishes for preparation and the decision to rejoin the droppers batch for JEE.

Approach to JEE Preparation

Students in the JEE droppers batch are encouraged to be aggressive in their preparation and dedicate ample time to study effectively.

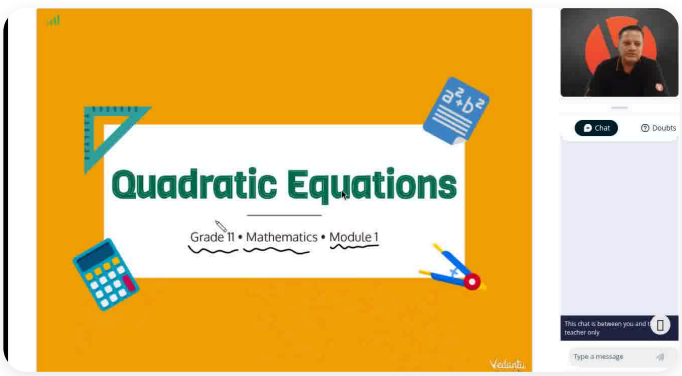
Feedback and Student Concerns

The instructor values student feedback and addresses concerns to ensure a beneficial learning experience. The teaching approach prioritizes what is best for students, rather than simply what they want.

Telegram Channel Resources

Students are encouraged to utilize the Telegram channel, where videos, assignments, and practice problems are posted regularly to aid in their progressive learning.

Introduction to Quadratic Equations and Syllabus Overview



particularly for JEE aspirants. Although this is a droppers' batch, the syllabus covered aligns with Grade 11 curriculum.

Key Topics to be Covered:

- Polynomial relations
- Relationship between roots and coefficients
- Nature of roots
- Transformations of equations
- Common roots
- Analysis of quadratic equations
- Location of roots (highly important for JEE Advanced)

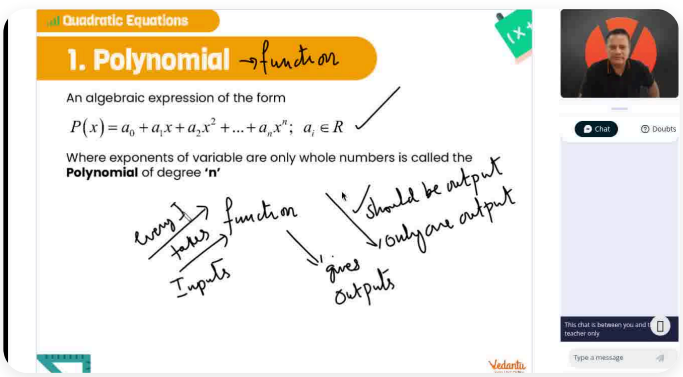
Importance for JEE:

While all topics are important for JEE Mains, JEE Advanced often integrates quadratic equations with other concepts. For instance, a 2025 JEE Advanced question combined quadratic equations with the binomial theorem. A strong foundation in the theory of equations is essential to tackle such integrated problems. Even cubic equations and their properties are relevant, as seen in a 2025 JEE Advanced question that depended on the graph of a cubic function.

Session Expectations:

We will cover numerous concepts, definitions, and formulas, and solve many problems within the allotted time. Students are encouraged to review the material before each session to maximize learning.

Introduction to Polynomials and Functions



What is a Polynomial?

An algebraic expression of the form $P(x) = a_0 + a_1x + a_2x^2 + ... + a_nx^n$, where $a_i \in R$, is called a **Polynomial of degree 'n'**. The exponents of the variable must be only whole numbers.

function is anything that takes inputs and gives outputs. For every input, there must be exactly one output.

- **Inputs:** These are the values taken by the function, collectively forming the **domain**.
- **Outputs:** These are the values given by the function, collectively forming the **range**.

If an input does not produce an output, or produces more than one output, then it is not considered a function; it is merely a relation.

Domain and Range of Polynomials

- **Domain:** For all polynomial functions, the domain is **R** (all real numbers). This means you can provide any real number as an input.
- **Range:** The range varies depending on the type of polynomial (linear, quadratic, cubic, etc.). Understanding the range for each type is crucial.

Key Terms

- **Variable:** Represented by 'x' in the general form.
- **Leading Coefficient:** The coefficient of the highest power of x (e.g., a_n in $a_n x^n$).
- **Degree:** The highest power of the variable 'n' in the polynomial. For example:
  - **Linear Polynomial:** Degree 1
  - **Quadratic Polynomial:** Degree 2
  - **Cubic Polynomial:** Degree 3

Identifying Polynomials

Condition for a Polynomial

An expression is a polynomial if and only if all the powers of the variable are **whole numbers** (0, 1, 2, 3, ...).

Example: Which of the following is a polynomial?

Let's consider some examples to identify polynomials:

1. **Expression A:** $x^{-1} + 2x + 1$
   - **Analysis:** The power of x in the first term is -1, which is not a whole number.
   - **Conclusion:** Not a polynomial.
2. **Expression B:** $\sqrt{x} + 3x^2 - 5$

3. **Expression C:** $\frac{x}{2} + 7$
    - **Analysis:** This can be written as $\frac{1}{2}x^1 + 7x^0$. The highest power of x is 1, which is a whole number.
    - **Conclusion:** This is a polynomial (specifically, a linear polynomial).
4. **Expression D:** $x^2 + \frac{1}{x}$
    - **Analysis:** $\frac{1}{x}$ can be written as x^{-1}. The power -1 is not a whole number.
    - **Conclusion:** Not a polynomial.

Therefore, **Option C** is the correct answer because the highest power (degree) is 1, which is a whole number.

Remainder Theorem Introduction

Definition

If a polynomial $P(x)$ is divided by a linear factor $(x - a)$, the remainder is $P(a)$. This means, to find the remainder, substitute the value 'a' (obtained by equating the linear factor to zero) into the polynomial $P(x)$.

Factor Theorem

If the remainder $P(a)$ is 0, then $(x - a)$ is a factor of the polynomial $P(x)$. Consequently, $x = a$ is a zero, root, or solution of the polynomial. For polynomials, the term "zero" is commonly used to denote a root, as it is the value of x for which $P(x) = 0$.

Applying Remainder Theorem: Example 1

Problem Statement

Find the remainder when $P(x) = x^3 + 2x^2 - x - 3$ is divided by $(x - 1)$.

Step-by-Step Solution

1. **Equate the divisor to zero:** Set $(x - 1) = 0$, which gives $x = 1$.
2. **Substitute the value of x into the polynomial:** Calculate $P(1)$. $P(1) = (1)^3 + 2(1)^2 - (1) - 3$
3. **Simplify the expression:** $P(1) = 1 + 2 - 1 - 3$ $P(1) = 3 - 1 - 3$ $P(1) = 2 - 3$ $P(1) = -1$

Conclusion

The remainder when $P(x)$ is divided by $(x - 1)$ is -1. This means that if you were to perform polynomial long division, the remainder would be -1. The Remainder Theorem provides a quicker

Notes

Problem Statement

Find the remainder when $P(x) = x^3 + 2x^2 - x - 3$ is divided by $(2x - 1)$.

Step-by-Step Solution

1. **Equate the divisor to zero:** Set $(2x - 1) = 0$, which gives $2x = 1$, so $x = \frac{1}{2}$.
2. **Substitute the value of x into the polynomial:** Calculate $P\left(\frac{1}{2}\right)$. $P\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^3 + 2\left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right) - 3$
3. **Simplify the expression:** $P\left(\frac{1}{2}\right) = \frac{1}{8} + 2\left(\frac{1}{4}\right) - \frac{1}{2} - 3$ $P\left(\frac{1}{2}\right) = \frac{1}{8} + \frac{2}{4} - \frac{1}{2} - 3$ $P\left(\frac{1}{2}\right) = \frac{1}{8} + \frac{1}{2} - \frac{1}{2} - 3$ The terms $\frac{1}{2}$ and $-\frac{1}{2}$ cancel out. $P\left(\frac{1}{2}\right) = \frac{1}{8} - 3$ To combine, find a common denominator: $P\left(\frac{1}{2}\right) = \frac{1}{8} - \frac{24}{8}$ $P\left(\frac{1}{2}\right) = -\frac{23}{8}$

Conclusion

The remainder when $P(x)$ is divided by $(2x - 1)$ is $-\frac{23}{8}$.

Understanding Factors and Zeros through Division

Polynomial Division and Remainder

When a polynomial $P(x)$ is divided by a linear factor $(x - a)$, if the remainder is 0, then $(x - a)$ is a factor of $P(x)$. This implies that $x = a$ is a root or zero of the polynomial.

Example: $x^2 - 3x + 2$ divided by $x - 1$

Let's demonstrate this with polynomial long division:

```
          x   - 2
       _____________
x - 1 | x^2 - 3x + 2
       - (x^2 - x)
       _____________
            -2x + 2
          - (-2x + 2)
          _____________
                   0
```

Analysis

- When $x^2 - 3x + 2$ is divided by $(x - 1)$, the remainder is 0.
- Therefore, $(x - 1)$ is a factor of $x^2 - 3x + 2$.

Notes

Problem: Finding p(3) given p(1)=0 and p(2)=0

Problem Statement

Find $p(3)$ if $p(1) = 0$ and $p(2) = 0$. Assume $p(x)$ is a quadratic polynomial of the form $x^2 + bx + c$.

Method 1: Using System of Equations

Given $p(x) = x^2 + bx + c$.

1. **Using $p(1) = 0$:** Substitute $x = 1$ into the polynomial: $1^2 + b(1) + c = 0$ $1 + b + c = 0$ (Equation 1)
2. **Using $p(2) = 0$:** Substitute $x = 2$ into the polynomial: $2^2 + b(2) + c = 0$ $4 + 2b + c = 0$ (Equation 2)
3. **Solving the System of Equations:** Subtract Equation 1 from Equation 2: $(4 + 2b + c) - (1 + b + c) = 0 - 0$ $4 + 2b + c - 1 - b - c = 0$ $3 + b = 0$ $b = -3$
4. **Finding the value of c:** Substitute $b = -3$ into Equation 1: $1 + (-3) + c = 0$ $-2 + c = 0$ $c = 2$
5. **Formulating the polynomial $p(x)$:** Now we have $b = -3$ and $c = 2$. So, the polynomial is: $p(x) = x^2 - 3x + 2$
6. **Calculating $p(3)$:** Substitute $x = 3$ into the polynomial: $p(3) = 3^2 - 3(3) + 2$ $p(3) = 9 - 9 + 2$ $p(3) = 2$

Method 2: Using Factor Theorem

1. **Understanding the Factor Theorem:** If $p(a) = 0$, then $(x - a)$ is a factor of the polynomial $p(x)$.
2. **Applying the Factor Theorem:** Since $p(1) = 0$, $(x - 1)$ is a factor of $p(x)$. Since $p(2) = 0$, $(x - 2)$ is a factor of $p(x)$.
3. **Constructing the polynomial:** As $p(x)$ is a quadratic polynomial and has factors $(x - 1)$ and $(x - 2)$, it can be written in the form: $p(x) = A(x - 1)(x - 2)$ Where A is the leading coefficient. From the problem statement, $p(x) = x^2 + bx + c$, which implies the leading coefficient $A = 1$. So, $p(x) = 1 \cdot (x - 1)(x - 2)$ $p(x) = (x - 1)(x - 2)$
4. **Expanding the polynomial (optional, for verification):** $p(x) = x^2 - 2x - x + 2$ $p(x) = x^2 - 3x + 2$ This matches the polynomial found in Method 1.

Both methods yield the same result, $p(3) = 2$.

Polynomial Roots and Factor Theorem

Understanding Polynomial Roots

If $P(x)$ is a polynomial, and $P(a) = 0$, then $x - a$ is a factor of $P(x)$. This is known as the Factor Theorem.

In this problem, we are given that $P(1) = 0$, $P(2) = 0$, and $P(3) = 0$. This implies:

- Since $P(1) = 0$, $(x - 1)$ is a factor.
- Since $P(2) = 0$, $(x - 2)$ is a factor.
- Since $P(3) = 0$, $(x - 3)$ is a factor.

Therefore, the polynomial $P(x)$ can be expressed as a product of these factors. If the leading coefficient is 1 (as implied by $1x^3 + bx^2 + cx + d$), then: $P(x) = (x - 1)(x - 2)(x - 3)$

Expanding the Product of Factors

To find the coefficients b, c, and d, we expand the product $(x - 1)(x - 2)(x - 3)$. This expansion can be done systematically using the relationships between roots and coefficients of a polynomial.

For a cubic polynomial $x^3 - S_1x^2 + S_2x - S_3 = 0$, where S_1, S_2, and S_3 are:

- S_1: Sum of the roots taken one at a time.
- S_2: Sum of the products of the roots taken two at a time.
- S_3: Product of the roots taken three at a time.

Given roots are $1, 2, 3$.

1. **Sum of roots (S_1):** $S_1 = 1 + 2 + 3 = 6$
2. **Sum of products of roots taken two at a time (S_2):** $S_2 = (1 \times 2) + (1 \times 3) + (2 \times 3)$ $S_2 = 2 + 3 + 6 = 11$
3. **Product of roots (S_3):** $S_3 = 1 \times 2 \times 3 = 6$

Substituting these values into the general cubic polynomial form: $P(x) = x^3 - (6)x^2 + (11)x - (6)$ $P(x) = x^3 - 6x^2 + 11x - 6$

Comparing Coefficients

Comparing this with the given polynomial form $P(x) = 1x^3 + bx^2 + cx + d$, we can identify the coefficients:

- $d = -6$

This method is more efficient than solving a system of linear equations derived from $P(1) = 0$, $P(2) = 0$, and $P(3) = 0$.

Introduction to Quadratic Equations

Definition of a Quadratic Equation

A quadratic equation is an equation of the form: $ax^2 + bx + c = 0$ Where:

- $a \neq 0$ (If $a = 0$, it becomes a linear equation).
- a is called the leading coefficient.
- a, b, and c can be real numbers or complex numbers.

Roots of a Quadratic Equation

A root of an equation is a value that, when substituted into the equation, makes the equation true (i.e., results in 0).

Number of Roots for a Quadratic Equation

A quadratic equation can have:

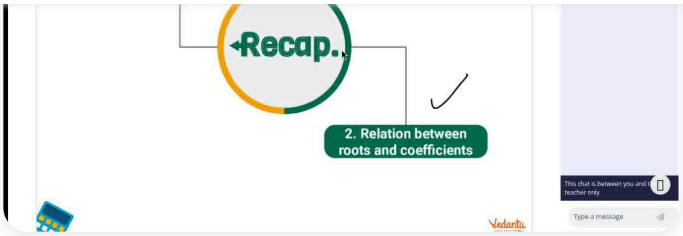
- **Zero real roots:** This implies two complex roots.
- **One real root:** This means two equal real roots.
- **Two real roots:** These can be distinct real roots.

Formula for Finding Roots

The roots of a quadratic equation can be found using the quadratic formula: $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

Discriminant

The term $b^2 - 4ac$ is called the **discriminant** and is denoted by Δ (or D). It determines the nature of the roots.



Definition of a Quadratic Equation

A quadratic equation is a polynomial equation of the second degree. It is generally represented in the form:

$ax^2 + bx + c = 0$

Where:

- a, b, and c are coefficients.
- $a \neq 0$ (If $a = 0$, the equation becomes linear, not quadratic).

Roots of a Quadratic Equation

A quadratic equation has at most two roots. These roots can be:

- Zero real roots
- One real root (a repeated root)
- Two distinct real roots

What is a Root?

A root (or solution) of a quadratic equation is a value, say α, such that when substituted into the equation, it makes the equation true (i.e., $a\alpha^2 + b\alpha + c = 0$).

Quadratic Formula

The roots of a quadratic equation $ax^2 + bx + c = 0$ can be found using the quadratic formula:

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Relation Between Roots and Coefficients

Let the two roots of the quadratic equation be α and β.

- **Sum of the roots (S_1):** The sum of the roots is given by: $\alpha + \beta = -\frac{b}{a}$
- **Product of the roots (S_2):** The product of the roots is given by: $\alpha \cdot \beta = \frac{c}{a}$

Notes

Relation Between Roots and Coefficients

Let α and β be the roots of the quadratic equation $ax^2 + bx + c = 0$.

Sum of the Roots (S_1)

The sum of the roots, denoted as S_1 (or simply S in lower classes), is given by: $S_1 = \alpha + \beta = -\frac{b}{a}$

Product of the Roots (S_2)

The product of the roots, denoted as S_2 (or simply P in lower classes), is given by: $S_2 = \alpha\beta = \frac{c}{a}$

Difference of the Roots

The difference of the roots, $|\alpha - \beta|$, can be derived using algebraic identities. We know that: $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$ Substituting the sum and product of roots: $(\alpha - \beta)^2 = \left(-\frac{b}{a}\right)^2 - 4\left(\frac{c}{a}\right)$ $(\alpha - \beta)^2 = \frac{b^2}{a^2} - \frac{4c}{a}$ $(\alpha - \beta)^2 = \frac{b^2-4ac}{a^2}$ Therefore, the difference of the roots is: $|\alpha - \beta| = \frac{\sqrt{b^2-4ac}}{|a|} = \frac{\sqrt{D}}{|a|}$

Generalization for Sums of Roots

In higher mathematics, the notation S_k is used to represent the sum of roots taken k at a time:

- S_1: Sum of the roots taking one at a time (i.e., $\alpha + \beta$).
- S_2: Sum of the roots taking two at a time (i.e., $\alpha\beta$).

Important Algebraic Identities

While specific formulas for expressions like $\alpha^3 + \beta^3$ or $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ are not explicitly memorized, they can be derived using basic algebraic identities such as:

- $(a + b)^2 = a^2 + b^2 + 2ab$
- $(a - b)^2 = a^2 + b^2 - 2ab$
- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Reciprocal Roots and Quadratic Equations

Understanding Reciprocal Roots

When dealing with quadratic equations, if one root is α and the other root is its reciprocal, $1/\alpha$, then a specific relationship exists between the coefficients of the quadratic equation.

need to determine the value of 'P' in the equation. The problem implies that the roots are reciprocal to each other, meaning $\alpha \neq 0$.

Key Formulas for Quadratic Equations

For a standard quadratic equation $Ax^2 + Bx + C = 0$, the relationships between the roots (let's say α and β) and the coefficients are:

- **Sum of roots:** $\alpha + \beta = -B/A$
- **Product of roots:** $\alpha \cdot \beta = C/A$

Applying the Product of Roots Formula

Given that the roots are α and $1/\alpha$, we can use the product of roots formula: $\alpha \cdot \left(\frac{1}{\alpha}\right) = \frac{C}{A}$ Simplifying the left side: $1 = \frac{C}{A}$ This implies that $C = A$.

Solving for P

In the context of the problem, the coefficients are represented as 'a', 'b', and 'c' (or 'P' for one of them). If the quadratic equation is structured such that 'c' is the constant term and 'a' is the coefficient of x^2, then the condition $C = A$ directly translates to $P = 2$ (assuming 'P' is one of these coefficients and '2' is the other, as implied by the solution $P = 2$).

Conclusion

If the roots of a quadratic equation are reciprocal to each other, then the coefficient of the x^2 term is equal to the constant term. In this specific problem, this leads to the conclusion that $P = 2$.

Understanding Symmetric Sums of Roots

Introduction to Symmetric Sums

In the context of quadratic equations, if α and β are the roots, we often encounter expressions involving powers of α and β. These are known as symmetric sums of roots.

Standard Notations

To simplify calculations, we use standard notations for the sum and product of roots:

- **Sum of roots (S1):** $\alpha + \beta = -\frac{b}{a}$
- **Product of roots (S2):** $\alpha\beta = \frac{c}{a}$

Deriving Common Symmetric Sums

- We know the algebraic identity: $a^2 + b^2 = (a + b)^2 - 2ab$
- Substituting α for a and β for b: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
- Using our standard notations: $\alpha^2 + \beta^2 = S1^2 - 2S2$
- Substituting the values of S1 and S2: $\alpha^2 + \beta^2 = \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2-2ac}{a^2}$

2. $\alpha^3 + \beta^3$
   - We know the algebraic identity: $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$
   - Substituting α for a and β for b: $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
   - Using our standard notations: $\alpha^3 + \beta^3 = S1^3 - 3S2 \cdot S1$
   - Substituting the values of S1 and S2: $\alpha^3 + \beta^3 = \left(-\frac{b}{a}\right)^3 - 3\left(\frac{c}{a}\right)\left(-\frac{b}{a}\right) = -\frac{b^3}{a^3} + \frac{3bc}{a^2} = \frac{-b^3+3abc}{a^3}$

Importance for JEE Examination

These derivations are crucial for the JEE examination. Students should be prepared for at least one question based on these concepts. Understanding how to express symmetric sums in terms of S1 and S2 simplifies problem-solving significantly.

Simplifying Expressions with Roots of a Quadratic Equation

Problem Statement

Given that α and β are the roots of a quadratic equation, and we define:

- $S_1 = \alpha + \beta$
- $S_2 = \alpha \cdot \beta$

We need to simplify the expression: $\alpha^{2/3} \cdot \alpha^{1/3} + \beta^{2/3} \cdot \beta^{1/3}/(\alpha^{1/3} \cdot \beta^{1/3})$

Solution Steps

1. **Identify the given expression:** The expression to simplify is $\alpha^{2/3} \cdot \alpha^{1/3} + \beta^{2/3} \cdot \beta^{1/3}/(\alpha^{1/3} \cdot \beta^{1/3})$.

2. **Simplify the terms in the numerator:** Using the exponent rule $a^m \cdot a^n = a^{m+n}$:
   - $\alpha^{2/3} \cdot \alpha^{1/3} = \alpha^{(2/3)+(1/3)} = \alpha^{3/3} = \alpha^1 = \alpha$
   - $\beta^{2/3} \cdot \beta^{1/3} = \beta^{(2/3)+(1/3)} = \beta^{3/3} = \beta^1 = \beta$

   So, the numerator simplifies to $\alpha + \beta$.

4. **Substitute the simplified terms back into the expression:** The expression becomes: $\frac{\alpha+\beta}{(\alpha\cdot\beta)^{1/3}}$
5. **Relate to S_1 and S_2:** We are given $S_1 = \alpha + \beta$ and $S_2 = \alpha \cdot \beta$. Substitute these into the simplified expression: $\frac{S_1}{(S_2)^{1/3}}$

Final Answer

The simplified value of the expression is $\frac{S_1}{(S_2)^{1/3}}$. To get a numerical answer, one would substitute the specific values of S_1 and S_2 (which are typically derived from the coefficients of the quadratic equation, e.g., $S_1 = -b/a$ and $S_2 = c/a$ for $ax^2 + bx + c = 0$).

Recommended Study Material and Strategy

Recommended Study Material

- **Vedantu Materials:** Tattva books, VDPP assignments, and Ved homework are sufficient.
- **Avoid Other Books:** Do not use other books like Singage, Pearson, Harian, R.D. Sharma, or Bharati Bhavan due to time constraints.

Previous Year Questions (PYQs)

- **JEE Advanced:** Solve all PYQs from 1979 onwards.
- **JEE Mains (AIEEE):** Solve all PYQs from 2002 onwards.

Exam Strategy

- **JEE Mains:** Aim to correctly solve a minimum of 12 questions. This typically yields 48-52 marks out of 100, which can result in a percentile of 95-98 or higher.
- **JEE Advanced:** In each paper (typically 17 questions), correctly solving 6-8 questions without mistakes can assure a good rank.

Simplification Example and Importance of Basics

Simplification Problem

Let's consider a simplification problem involving exponents. The problem involves terms like $\alpha^{2/3} \cdot \alpha^{1/3}$ and $\beta^{2/3} \cdot \beta^{1/3}$.

Step-by-Step Simplification

1. **Combine terms with the same base:**
    - For α: $\alpha^{2/3} \cdot \alpha^{1/3} = \alpha^{(2/3+1/3)} = \alpha^{3/3} = \alpha^1 = \alpha$

$\beta)^{1/3}$, it simplifies to $(\alpha + \beta)/(\alpha \cdot \beta)^{1/3}$.

Importance of Basic Concepts

- These simplifications are fundamental and are typically covered in 6th or 7th grade mathematics.
- A strong understanding of basic exponent rules is crucial for solving more complex problems in JEE.

Forming a Quadratic Equation from its Roots

Introduction

If α and β are the roots of a quadratic equation, we can construct the equation itself. The fundamental principle is that if $x = \alpha$ is a root, then $(x - \alpha)$ is a factor. Similarly, if $x = \beta$ is a root, then $(x - \beta)$ is a factor.

Derivation of the Equation

Since it's a quadratic equation, it will have exactly two factors. Multiplying these factors gives us the quadratic expression:

$$(x - \alpha)(x - \beta) = 0$$

Expanding this expression, we get:

$$x^2 - \beta x - \alpha x + \alpha\beta = 0 \; x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

Standard Form

This can be written in a more general form using the sum and product of roots:

$$x^2 - (\text{Sum of Roots})x + (\text{Product of Roots}) = 0$$

Let S_1 represent the sum of the roots and S_2 represent the product of the roots. Then the equation becomes:

$$x^2 - S_1 x + S_2 = 0$$

Where:

- $S_1 = \alpha + \beta$
- $S_2 = \alpha\beta$

Example Application

1. **Identify the new roots:** Let the new roots be α' and β'. In this case, $\alpha' = \frac{1}{\alpha}$ and $\beta' = \frac{1}{\beta}$.
2. **Calculate the sum of the new roots (S_1'):** $S_1' = \alpha' + \beta' = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta+\alpha}{\alpha\beta} = \frac{S_1}{S_2}$
3. **Calculate the product of the new roots (S_2'):** $S_2' = \alpha'\beta' = \left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right) = \frac{1}{\alpha\beta} = \frac{1}{S_2}$
4. **Form the new quadratic equation:** $x^2 - S_1'x + S_2' = 0$ Substituting the calculated values: $x^2 - \left(\frac{S_1}{S_2}\right)x + \left(\frac{1}{S_2}\right) = 0$

This method allows us to find the new quadratic equation by simply substituting the sum and product of the original roots (S_1 and S_2) into the expressions for the new sum and product of roots (S_1' and S_2').

Key Takeaway

The core concept is to always use the formula $x^2 - (\text{Sum of Roots})x + (\text{Product of Roots}) = 0$ when the roots are known. The complexity often lies in simplifying the expressions for the new sum and product of roots, but the fundamental approach remains consistent.

If the roots of a quadratic equation are known, say α and β, the quadratic equation can be constructed using the following formula:

$$x^2 - (\text{sum of the roots})x + (\text{product of the roots}) = 0$$

Or, using the notation for sum (S_1) and product (S_2) of roots:

$$x^2 - S_1x + S_2 = 0$$

Forming a Quadratic Equation from Roots

Problem Statement

Let p and q be two non-zero real numbers, and α and β are two numbers such that $\alpha^3 + \beta^3 = -p$ and $\alpha\beta = q$. We need to find the quadratic equation whose roots are $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$.

Key Concept

The general form of a quadratic equation whose roots are r_1 and r_2 is given by: $x^2 - (r_1 + r_2)x + (r_1r_2) = 0$ This can also be written as: $x^2 - (\text{Sum of the roots})x + (\text{Product of the roots}) = 0$

Given Information

- $\alpha^3 + \beta^3 = -p$
- $\alpha\beta = q$

2. **Calculate the Sum of the Roots (S):** $S = r_1 + r_2 = \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ To add these fractions, find a common denominator: $S = \frac{\alpha^2 \cdot \alpha + \beta^2 \cdot \beta}{\alpha\beta} = \frac{\alpha^3 + \beta^3}{\alpha\beta}$ Substitute the given values: $S = \frac{-p}{q}$
3. **Calculate the Product of the Roots (P):** $P = r_1 \cdot r_2 = \left(\frac{\alpha^2}{\beta}\right) \cdot \left(\frac{\beta^2}{\alpha}\right)$ Simplify the expression: $P = \frac{\alpha^2\beta^2}{\beta\alpha} = \alpha\beta$ Substitute the given value: $P = q$
4. **Form the Quadratic Equation:** Using the formula $x^2 - Sx + P = 0$, substitute the calculated values of S and P: $x^2 - \left(\frac{-p}{q}\right)x + q = 0$ $x^2 + \frac{p}{q}x + q = 0$
5. **Eliminate the denominator (optional, but common practice):** Multiply the entire equation by q (since $q \neq 0$): $qx^2 + px + q^2 = 0$

Final Answer

The quadratic equation whose roots are $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$ is $qx^2 + px + q^2 = 0$.

Quadratic Equations: Sum and Product of Roots

Understanding Roots and Coefficients

For a quadratic equation of the form $ax^2 + bx + c = 0$, where α and β are its roots:

- **Sum of the roots (S1):** $\alpha + \beta = -\frac{b}{a}$
- **Product of the roots (S2):** $\alpha\beta = \frac{c}{a}$

These relationships are fundamental for solving problems involving quadratic equations without explicitly finding the roots.

Quiz Problem: Forming a New Quadratic Equation

Consider a quadratic equation with roots α and β. We need to form a new quadratic equation whose roots are $\frac{1}{\alpha+1}$ and $\frac{1}{\beta+1}$.

Steps to Solve:

1. **Identify the sum and product of the original roots:** Use the formulas $S_1 = \alpha + \beta = -\frac{b}{a}$ and $S_2 = \alpha\beta = \frac{c}{a}$.
2. **Calculate the sum of the new roots:** Let the new roots be α' and β'. $\alpha' + \beta' = \frac{1}{\alpha+1} + \frac{1}{\beta+1}$ Combine the fractions: $\frac{(\beta+1)+(\alpha+1)}{(\alpha+1)(\beta+1)} = \frac{\alpha+\beta+2}{\alpha\beta+\alpha+\beta+1}$ Substitute the values of $\alpha + \beta$ and $\alpha\beta$ from the original equation.

4. **Form the new quadratic equation:** The general form of a quadratic equation with roots α' and β' is: $x^2 - (\alpha' + \beta')x + (\alpha'\beta') = 0$ Substitute the calculated sum and product of the new roots into this equation.

Importance of Practice and Hard Work

Success in competitive exams like JEE relies heavily on consistent hard work, strategic preparation, and smart study techniques rather than just innate intelligence. It is crucial to practice a wide variety of problems, understand the concepts thoroughly, and not hesitate to seek clarification for any difficulties encountered, especially with problem-solving approaches, techniques, or shortcuts.

Understanding the Original Quadratic Equation

Identifying the Roots and Equation

If α and β are the roots of a quadratic equation, then substituting α or β into the equation will satisfy it. The given quadratic equation is $x^2 - 5x + 3 = 0$.

Sum and Product of Roots

For a quadratic equation $ax^2 + bx + c = 0$, the sum of the roots (S1) is given by $-b/a$, and the product of the roots (S2) is given by c/a.

In our case, for $x^2 - 5x + 3 = 0$:

- Sum of roots (S1): $\alpha + \beta = -(-5)/1 = 5$
- Product of roots (S2): $\alpha\beta = 3/1 = 3$

This is a common problem found in many textbooks, where α and β satisfy the equation, confirming they are its roots.

Forming a New Quadratic Equation with Transformed Roots

New Roots

We need to form a new quadratic equation whose roots are α/β and β/α.

General Form of a Quadratic Equation

The general form of a quadratic equation with roots r_1 and r_2 is $x^2 - (r_1 + r_2)x + (r_1 r_2) = 0$.

Sum of the New Roots

Let the new roots be $r_1 = \alpha/\beta$ and $r_2 = \beta/\alpha$.

We know that $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$. Using the values from the original equation:

- $\alpha + \beta = S1 = 5$
- $\alpha\beta = S2 = 3$

So, $\alpha^2 + \beta^2 = (5)^2 - 2(3) = 25 - 6 = 19$.

Now, substitute this back into the sum of new roots: Sum of new roots = $\frac{19}{3}$.

Product of the New Roots

Product of new roots: $r_1 r_2 = \left(\frac{\alpha}{\beta}\right)\left(\frac{\beta}{\alpha}\right) = 1$.

Constructing the New Equation

Substitute the sum and product of the new roots into the general form: $x^2 - \left(\frac{19}{3}\right)x + 1 = 0$

To eliminate the fraction, multiply the entire equation by 3: $3x^2 - 19x + 3 = 0$

This is the new quadratic equation with roots α/β and β/α.

Difference Between Roots of Quadratic Equations

Problem Statement

If $A \boxminus B$, and the difference between the roots of two quadratic equations is the same, we need to find the relationship between their coefficients.

Let the roots of the first equation be α and β. Let the roots of the second equation be γ and δ.

We are given that the difference between the roots is the same: $\alpha - \beta = \gamma - \delta$

Approach to Solving the Problem

To work with the difference of roots, it's often easier to square the expression, as we know the relationship between the sum and product of roots and the square of their difference.

Squaring both sides of the given condition: $(\alpha - \beta)^2 = (\gamma - \delta)^2$

We know the identity: $(a - b)^2 = (a + b)^2 - 4ab$. Applying this identity to both sides: $(\alpha + \beta)^2 - 4\alpha\beta = (\gamma + \delta)^2 - 4\gamma\delta$

Relating to Sum and Product of Roots

Let's consider two general quadratic equations:

- Product of roots (S_2): $\alpha\beta = B$

2. For the second equation, let its roots be γ and δ.
   - Sum of roots (S_1'): $\gamma + \delta = -B$
   - Product of roots (S_2'): $\gamma\delta = A$

Substitution and Simplification

Substitute these values into the squared difference equation: $(-A)^2 - 4(B) = (-B)^2 - 4(A)$
$A^2 - 4B = B^2 - 4A$

Rearrange the terms to find the relationship: $A^2 - B^2 = 4B - 4A$ $(A - B)(A + B) = -4(A - B)$

Since we are given that $A \neq B$, we can divide both sides by $(A - B)$: $A + B = -4$

This is the final relationship between the coefficients A and B given the condition.

Homework and Study Strategy

Homework Guidelines

- **Advanced Problems:** Do not attempt advanced problems until instructed.
- **PYQs (Previous Year Questions):** Start with 2026 PYQs, specifically focusing on quadratic equations.
  - **Source for PYQs:** Download 2026 papers or use websites like MathonGo, which provides monthly, yearly, and lesson-wise PYQs.

Addressing Time Constraints and Learning Pace

- If you have limited time or a slower learning methodology, restrict your practice to only PYQs.

Performance in Weekend Tests

- **Strategy for Good Marks:** To score well in weekend tests, ensure you:
  - Listen attentively to sessions.
  - Solve quiz problems.
  - Complete daily homework (10-20 questions).
- **Expected Score:** Following this strategy should help you achieve 40-50% marks.

Mathematics in JEE

lengthy. **Lengthy Nature:** Be prepared for mathematics sections in both JEE Mains and Advanced to be lengthy.

Quadratic Equations - Solution Approach

Problem-Solving Example (Conceptual Discussion)

- The lecturer briefly discusses a solution approach, mentioning variables like S_1, S_2, $S_1^{'}$, and $S_2^{'}$.
- **Variables Mentioned:**
  - $S_1 = -A$
  - $S_2 = B$
  - $S_1^{'} = -b/a$
  - $S_2^{'} = c/a$
- **Given Condition:** It is given that $a \boxminus b$.
- **Simplification:** The problem involves simplifications based on the condition $a + b$.
- **Concept:** The underlying concept is described as "very easy."

Importance of Quadratic Equations

- Quadratic equations should generally be an easy topic for students.
- **JEE Advanced Context:** In JEE Advanced, quadratic equations are often mixed with concepts from other chapters, requiring integrated problem-solving skills.

Problem: Quadratic Equation with Consecutive Odd Integer Roots

Problem Statement

If the roots of the quadratic equation $x^2 - ax + b = 0$ are two consecutive odd integers, then find the value of $a^2 - 4b$.

Understanding Consecutive Odd Integers

Consecutive odd integers can be represented in a few ways:

- $2n - 1$ and $2n + 1$
- $2n + 1$ and $2n + 3$

Method 1: Direct Substitution (Normal Student Method)

Let the roots of the quadratic equation be α and β.

- Sum of roots: $\alpha + \beta = -(-a)/1 = a$
- Product of roots: $\alpha\beta = b/1 = b$

Let's choose the roots as $\alpha = 2n - 1$ and $\beta = 2n + 1$.

1. **Calculate the sum of the roots:** $\alpha + \beta = (2n - 1) + (2n + 1) = 4n$ Therefore, $a = 4n$.
2. **Calculate the product of the roots:** $\alpha\beta = (2n - 1)(2n + 1) = (2n)^2 - 1^2 = 4n^2 - 1$ Therefore, $b = 4n^2 - 1$.
3. **Substitute 'a' and 'b' into the expression $a^2 - 4b$:** $a^2 - 4b = (4n)^2 - 4(4n^2 - 1) = 16n^2 - (16n^2 - 4) = 16n^2 - 16n^2 + 4 = 4$

Thus, $a^2 - 4b = 4$.

Method 2: Using the Difference of Roots (Smarter Method)

This method utilizes the property that the difference between any two consecutive odd integers is 2.

Let the roots be α and β. We are given that α and β are consecutive odd integers, so their difference is 2. $|\alpha - \beta| = 2$

Squaring both sides: $(\alpha - \beta)^2 = 2^2 = 4$

We know the identity relating the difference of roots to the sum and product of roots: $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

From the quadratic equation $x^2 - ax + b = 0$:

- $\alpha + \beta = a$
- $\alpha\beta = b$

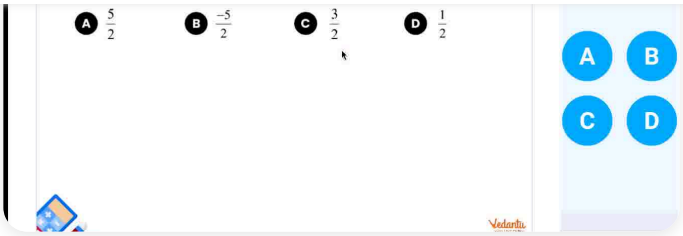
Substitute these into the identity: $(\alpha - \beta)^2 = (a)^2 - 4(b)$

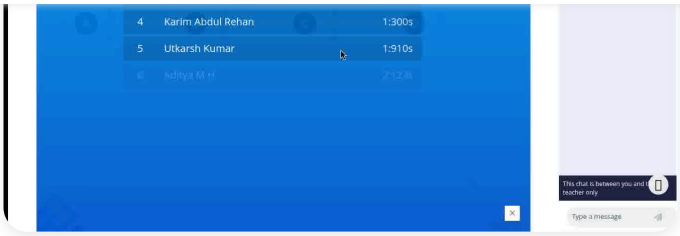
Since $(\alpha - \beta)^2 = 4$, we have: $a^2 - 4b = 4$

Both methods yield the same result, $a^2 - 4b = 4$. The second method is quicker if one recognizes the property of the difference of roots.

Quiz: Product and Sum of Roots

Notes





Problem Statement

If the product of roots of the equation $(a + 1)x^2 + (2a + 3)x + 4 = 0$ is 2, then the sum of roots is: A. $\frac{5}{2}$ B. $-\frac{5}{2}$ C. $\frac{3}{2}$ D. $\frac{1}{2}$

Solution

1. **Identify the coefficients:** For a quadratic equation $Ax^2 + Bx + C = 0$, the coefficients are: $A = (a + 1)$ $B = (2a + 3)$ $C = 4$
2. **Use the product of roots formula:** The product of the roots ($\alpha\beta$) is given by $\frac{C}{A}$. We are given that the product of the roots is 2. $\frac{C}{A} = 2$ $\frac{4}{a+1} = 2$
3. **Solve for 'a':** Multiply both sides by $(a + 1)$: $4 = 2(a + 1)$ Distribute the 2: $4 = 2a + 2$ Subtract 2 from both sides: $2 = 2a$ Divide by 2: $a = 1$
4. **Substitute 'a' back into the original equation:** Substitute $a = 1$ into the given equation $(a + 1)x^2 + (2a + 3)x + 4 = 0$: $(1 + 1)x^2 + (2(1) + 3)x + 4 = 0$ $2x^2 + (2 + 3)x + 4 = 0$ $2x^2 + 5x + 4 = 0$
5. **Calculate the sum of roots:** The sum of the roots ($\alpha + \beta$) is given by $-\frac{B}{A}$. From the modified equation $2x^2 + 5x + 4 = 0$, we have: $A = 2$ $B = 5$ $C = 4$ Therefore, the sum of the roots is: $\alpha + \beta = -\frac{5}{2}$

Final Answer

The sum of the roots is $-\frac{5}{2}$. This corresponds to option B.

General Formula for Sum of Roots (SR)

Introduction to SR

In the study of polynomial equations, understanding the relationship between the roots and the coefficients is crucial. For a polynomial of degree N, we define S_R as the sum of the roots taken R at a time. This means we select R roots, multiply them together, and then sum up all such possible products.

General Formula

- N is the degree of the polynomial (the highest power of X).
- R is the number of roots taken at a time (e.g., for S_1, $R = 1$; for S_2, $R = 2$; for S_3, $R = 3$, and so on).
- The term "Coefficient of X^{N-R}" refers to the coefficient of the term whose power is $N - R$.
- The term "Coefficient of X^N" refers to the coefficient of the leading term (the term with the highest power of X).

Understanding S_R

- S_1 **(Sum of roots taken one at a time):** This is simply the sum of all individual roots. For a general polynomial $a_N X^N + a_{N-1} X^{N-1} + \cdots + a_1 X + a_0 = 0$, $S_1 = -\frac{a_{N-1}}{a_N}$.
- S_2 **(Sum of products of roots taken two at a time):** This involves taking all possible pairs of roots, multiplying each pair, and then summing these products. For example, if the roots are α, β, γ, then $S_2 = \alpha\beta + \beta\gamma + \gamma\alpha$.
- S_3 **(Sum of products of roots taken three at a time):** Similar to S_2, but with groups of three roots. For roots $\alpha, \beta, \gamma, \delta$, $S_3 = \alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$.

Application to Specific Cases

- **Cubic Equation:** For a cubic equation $aX^3 + bX^2 + cX + d = 0$, $N = 3$.
  - $S_1 = (-1)^1 \frac{\text{Coefficient of } X^{3-1}}{\text{Coefficient of } X^3} = -\frac{b}{a}$
  - $S_2 = (-1)^2 \frac{\text{Coefficient of } X^{3-2}}{\text{Coefficient of } X^3} = +\frac{c}{a}$
  - $S_3 = (-1)^3 \frac{\text{Coefficient of } X^{3-3}}{\text{Coefficient of } X^3} = -\frac{d}{a}$ (This is the product of the roots).
- **Quadratic Equation:** For a quadratic equation $aX^2 + bX + c = 0$, $N = 2$.
  - $S_1 = (-1)^1 \frac{\text{Coefficient of } X^{2-1}}{\text{Coefficient of } X^2} = -\frac{b}{a}$
  - $S_2 = (-1)^2 \frac{\text{Coefficient of } X^{2-2}}{\text{Coefficient of } X^2} = +\frac{c}{a}$ (This is the product of the roots).

Alternating Signs

Notice the alternating signs in the formulas for S_R: S_1 is negative, S_2 is positive, S_3 is negative, and so on. This pattern is captured by the $(-1)^R$ term in the general formula.

Forming Polynomial Equations from Roots

General Formula for Cubic Equation

If the roots of a cubic equation are known, the equation can be formed using the following general formula:

- S_1 is the sum of the roots taken one at a time (i.e., $\alpha + \beta + \gamma$).
- S_2 is the sum of the products of the roots taken two at a time (i.e., $\alpha\beta + \beta\gamma + \gamma\alpha$).
- S_3 is the product of the roots taken three at a time (i.e., $\alpha\beta\gamma$).

Therefore, the cubic equation can also be written as:

$$x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma = 0$$

General Formula for Quartic Equation

For a fourth-degree (quartic) equation with roots, the formula follows a similar pattern with alternating signs:

$$x^4 - S_1x^3 + S_2x^2 - S_3x + S_4 = 0$$

Where:

- S_1: Sum of roots taken one at a time.
- S_2: Sum of products of roots taken two at a time.
- S_3: Sum of products of roots taken three at a time.
- S_4: Product of roots taken four at a time.

Alternative Method for Forming Equations

Another way to form a polynomial equation if its roots (e.g., $\alpha_1, \alpha_2, \ldots, \alpha_n$) are known is by using the product of linear factors:

$$(x - \alpha_1)(x - \alpha_2)\ldots(x - \alpha_n) = 0$$

For example:

- **For two roots (quadratic equation):** $(x - \alpha)(x - \beta) = 0$
- **For three roots (cubic equation):** $(x - \alpha)(x - \beta)(x - \gamma) = 0$

Next Steps and Study Resources

Module 2: Cubic Equation and Problem Solving

We have covered up to the cubic equation. The next session will focus on solving problems related to this topic. Remember, Module 2 will delve into the cubic equation, its relations, and problem-solving techniques.

haven't been covered yet, do not worry. Skip those problems and move on to the ones you can attempt. The goal is to try all problems you are capable of solving. Any problems you cannot solve will be addressed and clarified in subsequent sessions.

- **Homework Resources:**
  - VDDP books and Tatva books are available in the library. Students are encouraged to start using these resources immediately.
  - PYQs (Previous Year Questions) are also available.
  - Veda AI provides homework assignments.
  - Assignments, including quadratic equations, have been shared on the Telegram channel.

Exam Preparation and Mindset

- **JEE Advanced:** The JEE Advanced exam is approaching on May 17th. It is crucial to start studying diligently now.
- **Importance of Exams:** Do not skip exams. The focus is not solely on marks but on consistent effort and understanding. The key questions are:
  - "Are you present?" (Attending sessions)
  - "Did you do all the problems?" (Attempting assignments and practice)
- **Negative Marking:** Strive to solve problems without mistakes to avoid negative marking.
- **Support:** Solutions and clarifications will be provided. Patience is key.

Session Feedback and Future Improvements

- The instructor is committed to bringing about positive changes in the sessions based on understanding the batch dynamics and student needs. This is a gradual process.

Closing Remarks

Did you find these notes helpful?

- Thank you for attending the session. Please continue learning and make the best use of your time. Do not pressurize yourself excessively, but strive for your best within the available time. The session will now be closed as there is another session scheduled at 3:30.